

Exact Enumeration and Fixed-Rank Asymptotics of Ballot-Admissible Fibonacci Ribbon Tableaux

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Abstract

Tenn asked for the enumeration of ballot-admissible Fibonacci ribbon tableaux beyond the three-letter case. We solve this problem and simultaneously cover every alphabet size $n \geq 2$ by proving the following exact formula. If $b_{n,k}$ denotes the number of ballot-admissible $\text{Fib}(n, k)$ ribbons and f^λ is the number of standard Young tableaux of shape λ , then

$$b_{n,k} = \sum_{j=0}^{\lfloor k/2 \rfloor} (-1)^j \binom{k-j}{j} \sum_{\substack{\lambda \vdash k-2j \\ \ell(\lambda) \leq n}} f^\lambda.$$

The proof encodes the columns as a dominant type- A walk alternating between the defining representation and its dual. The unrestricted mixed walk has the bounded-height standard-tableau count, while each forbidden adjacency is a ballot-neutral block; endpoint-preserving contraction and inclusion–exclusion give both the formula and a highest-weight refinement in explicit Schur coefficients.

The resulting generating-function substitution, together with exact Weyl integral representations and Regev’s strip asymptotics, yields an explicit constant $K_n > 0$ such that, for every fixed $n \geq 3$,

$$b_{n,k} \sim K_n \left(\frac{n + \sqrt{n^2 - 4}}{2} \right)^k k^{-n(n-1)/4}.$$

For $n \geq k$, the count stabilizes to involutions with no adjacent transposition. Their number is asymptotic to e^{-1} times the involution number, proving a conjecture recorded in OEIS A170941. More generally, for each fixed r , the defect on the line $n = k - r$ is eventually a polynomial in k of degree $r - 1$ with leading coefficient $I_{r-1}/(r-1)!$, where I_m is the m th involution number.

1 Introduction

Fibonacci ribbons are alternating-height skew tableaux introduced by Donnelly, Dunkum, Lišková, and Nance in their study of symmetric Fibonacci distributive lattices and special-linear representations [2]. A ribbon is ballot-admissible when it is a Littlewood–Richardson tableau. For three letters, empirical data suggested the peakless Motzkin numbers. Tenn proved that conjecture

and constructed an explicit bijection, then left the enumeration of $\text{Fib}(n, k)$ for $n \geq 4$ as an open problem [9].

The main result of this paper is an exact finite formula for every n and k . The derivation uses two structural facts. First, if the single local semistandard obstruction is temporarily allowed, the alternating short and tall columns form a mixed defining/dual branching walk. Although the endpoint distributions depend on which factors are dual, their total multiplicity does not. Second, each forbidden adjacent pair reads exactly $12 \cdots n$, so it may be deleted without changing any ballot difference. The obstruction sets are matchings of a path, and inclusion–exclusion gives the answer.

The contraction preserves the final dominant weight, which yields a highest-weight refinement in explicit Schur coefficients. The ordinary generating function is obtained from the bounded-tableau series by a rational substitution. Combining this substitution with exact Weyl integral representations and Regev’s strip asymptotics gives, for every fixed $n \geq 3$, an explicit constant K_n such that

$$b_{n,k} \sim K_n \left(\frac{n + \sqrt{n^2 - 4}}{2} \right)^k k^{-n(n-1)/4}.$$

The stable range leads to a second theme. The sum of the numbers of standard tableaux of all shapes of size k is the involution number. The same path-matching inclusion–exclusion excludes the cycles $(i, i + 1)$. This identifies the stable ribbon numbers with OEIS A170941 and supplies a short Poisson-limit proof of the conjectural limiting proportion recorded there [7].

The approach to stability is also explicit. On every fixed-distance line $n = k - r$, the deficit from the stable count is a finite sum of bounded-tail tableau numbers and is eventually a polynomial in k of degree $r - 1$. Together with the fixed-rank asymptotic, this also shows that the two natural iterated density limits do not commute.

2 Fibonacci ribbons and column parameters

Fix integers $n \geq 2$ and $k \geq 0$. The Fibonacci ribbon shape has k columns. From right to left their heights alternate between 1 and $n - 1$, starting with height 1. Entries lie in $[n] = \{1, \dots, n\}$, columns are strictly increasing, and rows are weakly increasing. The linear word is read column by column from right to left and within a column from top to bottom. The tableau is ballot-admissible if every prefix w' of this word satisfies

$$\#_1(w') \geq \#_2(w') \geq \cdots \geq \#_n(w').$$

These are the conventions of [2, 9].

Parameterize the c th column from the right by $a_c \in [n]$. If c is odd, the column is the singleton a_c . If c is even, it contains every element of $[n]$ except

$$y_c = n + 1 - a_c,$$

in increasing order. The Fibonaccian-string condition from [2, Theorem 4.2], translated from left-to-right to right-to-left order, becomes

$$(a_c, a_{c+1}) \neq (1, n). \tag{1}$$

Thus the local obstruction is unique.

For a prefix w , define its difference vector

$$d(w) = (d_1, \dots, d_{n-1}), \quad d_i = \#_i(w) - \#_{i+1}(w).$$

The ballot condition is $d(w) \in \mathbb{Z}_{\geq 0}^{n-1}$ at every prefix. Put

$$v_1 = e_1, \quad v_x = -e_{x-1} + e_x \quad (1 < x < n), \quad v_n = -e_{n-1}.$$

Lemma 2.1 (Literal column walk). *Reading a singleton x changes d by v_x . Reading a tall column omitting y changes d by $-v_y$. In either case the column may be appended to a ballot prefix exactly when the resulting difference vector is nonnegative.*

Proof. The singleton statement follows directly from the definition of the differences. A full increasing block $12 \cdots n$ returns every difference to its initial value and stays nonnegative when started from a nonnegative vector. Omitting y therefore has net increment $-v_y$.

It remains to check internal prefixes. Before the omitted position, the increasing letters merely move one unit of surplus to the right. If $y < n$, the only possible new violation occurs when $y + 1$ is read, and it occurs exactly when the final coordinate required by $d - v_y \geq 0$ is absent. If $y = n$, there is no later letter and no additional condition. Hence endpoint nonnegativity is necessary and sufficient. $\square$

Let $b_{n,k}$ denote the desired ribbon count. Let $u_{n,k}$ denote the number obtained from the same ballot column words when Equation (1) is not imposed.

3 The unrestricted mixed branching walk

Let $D_n = \mathbb{Z}_{\geq 0}^{n-1}$. On functions on D_n , let A be the adjacency operator for the valid steps v_x , and let B be the adjacency operator for the valid steps $-v_y$. All sums below are finite at every state.

Lemma 3.1 (Mixed branching). *The operators A and B commute, and*

$$A\mathbf{1} = B\mathbf{1},$$

where $\mathbf{1}$ is the constant-one function on D_n .

Proof. The vectors $v_x - v_y$ are distinct when $x \neq y$. For such a pair, the validity conditions of the two steps are independent except when $x = y + 1$. In that exceptional case, both orders are valid exactly when $d_y \geq 2$: each step lowers the same boundary coordinate by one. Hence a step v_x followed by $-v_y$ is valid if and only if the reverse order is valid for every $x \neq y$, so the off-diagonal matrix entries of AB and BA agree.

For a zero-displacement two-step walk, the two letters must have $x = y$. The number of valid A -moves from d is

$$1 + \#\{i : d_i > 0\};$$

the always-valid move is $x = 1$. The number of valid B -moves is the same, with always-valid move $y = n$. Hence the diagonal entries also agree and $AB = BA$. The same outdegree calculation gives $A\mathbf{1} = B\mathbf{1}$. $\square$

Theorem 3.2 (Unrestricted count). *For every $n \geq 2$ and $k \geq 0$,*

$$u_{n,k} = \sum_{\substack{\lambda \vdash k \\ \ell(\lambda) \leq n}} f^\lambda, \tag{2}$$

where $f^\lambda = |\text{SYT}(\lambda)|$.

Proof. There are $a = \lceil k/2 \rceil$ defining steps and $c = \lfloor k/2 \rfloor$ dual steps. If δ_0 is evaluation at the zero weight, the total unrestricted count is

$$\delta_0 A^a B^c \mathbf{1}.$$

Lemma 3.1 gives, inductively, $B^r \mathbf{1} = A^r \mathbf{1}$: indeed

$$B^{r+1} \mathbf{1} = B A^r \mathbf{1} = A^r B \mathbf{1} = A^{r+1} \mathbf{1}.$$

Consequently $u_{n,k} = \delta_0 A^k \mathbf{1}$.

A path of k defining steps from zero is a word $x_1 \cdots x_k$ on $[n]$ whose prefix multiplicities are weakly decreasing. Place j in row x_j . Rows increase because positions are inserted in order, and the prefix inequalities say precisely that every entry has a cell above it when required. This is a standard Young tableau with at most n rows. Reading the row containing each successive entry reverses the construction. Summing over shapes proves Equation (2). $\square$

By the hook-length formula,

$$f^\lambda = \frac{k!}{\prod_{z \in \lambda} h(z)} \quad (\lambda \vdash k), \quad (3)$$

so Equation (2) is directly computable. Robinson–Schensted also identifies $u_{n,k}$ with the number of involutions of $[k]$ whose longest decreasing subsequence has length at most n .

4 Exact enumeration by neutral-block contraction

Lemma 4.1 (Neutral bad pairs). *Suppose a specified adjacency satisfies $(a_c, a_{c+1}) = (1, n)$. Its two columns have combined linear word $12 \cdots n$. Deleting the two columns is a bijection between unrestricted ballot column words having that specified bad pair and unrestricted ballot column words of length $k - 2$.*

More generally, if j pairwise disjoint adjacencies are specified to be bad, simultaneously deleting them is a bijection with unrestricted objects of length $k - 2j$.

Proof. If c is odd, the two blocks are the singleton 1 and the tall column $2 \cdots n$. If c is even, they are $1 \cdots (n-1)$ and the singleton n . Their concatenation is $12 \cdots n$ in both cases. This block returns every difference to its initial value, and every internal prefix is ballot. Deletion and insertion are inverse operations. Removing two columns preserves the parity of every later column, so the argument applies simultaneously to any disjoint collection. $\square$

Bad adjacencies cannot overlap, since a shared parameter would have to equal both 1 and n . Sets of j possible bad locations are therefore the j -edge matchings of a path on k vertices, of which there are

$$\binom{k-j}{j}. \quad (4)$$

Theorem 4.2 (Fibonacci-ribbon enumeration). *For every $n \geq 2$ and $k \geq 0$,*

$$b_{n,k} = \sum_{j=0}^{\lfloor k/2 \rfloor} (-1)^j \binom{k-j}{j} \sum_{\substack{\lambda \vdash k-2j \\ \ell(\lambda) \leq n}} f^\lambda. \quad (5)$$

Equations (3) and (5) give a finite explicit formula.

Proof. Apply inclusion–exclusion to the $k-1$ bad-adjacency events. An intersection is empty unless its locations form a matching. For a matching of size j , Lemma 4.1 identifies the intersection with the unrestricted objects counted by $u_{n,k-2j}$. Use Equations (4) and (2). $\square$

The same contraction retains the final dominant weight and therefore admits a highest-weight refinement. For a dominant difference vector $d = (d_1, \dots, d_{n-1})$, let

$$\mu_i = d_i + d_{i+1} + \dots + d_{n-1} \quad (1 \leq i < n), \quad \mu_n = 0.$$

Thus μ is the normalized partition representing the corresponding $\mathfrak{sl}_n$ highest weight. Let $b_{n,k}^\mu$ count admissible ribbons whose column walk ends at this weight. Put

$$a = \left\lceil \frac{k}{2} \right\rceil, \quad c = \left\lfloor \frac{k}{2} \right\rfloor.$$

For nonnegative r, s , define

$$M_{r,s}^{(n)}(\mu) = \begin{cases} \langle p_1^r e_{n-1}^s, s_{\mu+(q^n)} \rangle, & q = \frac{r + (n-1)s - |\mu|}{n} \in \mathbb{Z}_{\geq 0}, \\ 0, & \text{otherwise,} \end{cases} \quad (6)$$

where the bracket denotes the Schur coefficient, and $\mu + (q^n) = (\mu_1 + q, \dots, \mu_n + q)$.

Theorem 4.3 (Highest-weight refinement). *For every normalized dominant partition μ ,*

$$b_{n,k}^\mu = \sum_{j=0}^{\lfloor k/2 \rfloor} (-1)^j \binom{k-j}{j} M_{a-j, c-j}^{(n)}(\mu). \quad (7)$$

Summing Equation (7) over μ recovers Theorem 4.2.

Proof. Let V be the defining SL_n -module. As an SL_n -module, $V^* \cong \bigwedge^{n-1} V$. Hence the polynomial GL_n character whose restriction describes $V^{\otimes r} \otimes (V^*)^{\otimes s}$ is $p_1^r e_{n-1}^s$. A polynomial GL_n constituent $S_\lambda V$ restricts to normalized SL_n highest weight μ exactly when $\lambda = \mu + (q^n)$ for some $q \geq 0$. Homogeneous degree forces $|\lambda| = r + (n-1)s$, so q is the unique integer in Equation (6). The Pieri branching paths to μ are therefore counted by $M_{r,s}^{(n)}(\mu)$.

Every selected bad adjacency contains one defining and one dual column. Deleting its full-alphabet block changes (a, c) to $(a-1, c-1)$ but has zero difference displacement, so it preserves the final highest weight. The simultaneous contraction of j disjoint bad pairs therefore has endpoint multiplicity $M_{a-j, c-j}^{(n)}(\mu)$. Endpointwise inclusion–exclusion proves Equation (7). Summation over all endpoints is the unrestricted endpoint functional used in Theorem 3.2, which gives the last assertion. $\square$

The first rows, beginning at $k = 0$, are

$n \backslash k$	0	1	2	3	4	5	6	7	8	9	10	11	12
2	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	2	4	8	17	37	82	185	423	978	2283
4	1	1	1	2	5	12	31	82	229	646	1891	5574	16883
5	1	1	1	2	5	13	36	105	321	1018	3334	11216	38635
6	1	1	1	2	5	13	37	111	355	1179	4099	14665	54299

For $n = 3$, Theorem 4.2 recovers the peakless Motzkin numbers and therefore agrees with the bijection of [9].

5 Generating functions and finite-rank structure

Put

$$U_n(z) = \sum_{k \geq 0} u_{n,k} z^k, \quad B_n(z) = \sum_{k \geq 0} b_{n,k} z^k.$$

For fixed n these are convergent near the origin. They may also be read as formal power series.

Theorem 5.1 (Exact substitution). *For every $n \geq 2$,*

$$B_n(z) = \frac{1}{1+z^2} U_n\left(\frac{z}{1+z^2}\right). \quad (8)$$

Consequently $B_n(z)$ is D-finite for every fixed n .

Proof. In Theorem 4.2, put $m = k - 2j$ and sum over k . The elementary identity

$$\sum_{j \geq 0} \binom{m+j}{j} t^j = (1-t)^{-m-1}$$

gives

$$\begin{aligned} B_n(z) &= \sum_{m \geq 0} u_{n,m} z^m \sum_{j \geq 0} (-1)^j \binom{m+j}{j} z^{2j} \\ &= \frac{1}{1+z^2} \sum_{m \geq 0} u_{n,m} \left(\frac{z}{1+z^2}\right)^m. \end{aligned}$$

For fixed n , the bounded-height tableau sequence $u_{n,m}$ is P -recursive; equivalently its generating function is D-finite [1]. D-finite series are closed under multiplication by a rational function and algebraic substitution, proving the last assertion. $\square$

The substitution is invertible as a formal series. If

$$\psi(w) = \frac{1 - \sqrt{1 - 4w^2}}{2w} = w + w^3 + 2w^5 + \cdots,$$

then $\psi/(1+\psi^2) = w$ and

$$U_n(w) = (1 + \psi(w)^2) B_n(\psi(w)). \quad (9)$$

Thus the ribbon and bounded-involution enumerations determine one another.

For $n = 3$, $u_{3,k}$ is the Motzkin number, so $U_3 = 1 + zU_3 + z^2U_3^2$. Equation (8) simplifies to

$$B_3(z) = 1 + zB_3(z) + z^2B_3(z)(B_3(z) - 1),$$

which is exactly the peakless-Motzkin decomposition used in [9]. For $n = 2$, the same substitution reduces to $B_2(z) = 1/(1-z)$, agreeing with $b_{2,k} = 1$.

The finite-dimensional Bessel system remains useful for exact recurrences. Let

$$Y_n(x) = \sum_{k \geq 0} u_{n,k} \frac{x^k}{k!}, \quad J_r(x) = \sum_{m \geq 0} \frac{x^{2m+r}}{m!(m+r)!}.$$

Writing $d = \lfloor n/2 \rfloor$, Gessel's determinant in the simplified form of Bergeron and Gascon places Y_n in the $\mathbb{C}(x)$ -span

$$\{J_0^a J_1^{d-a} : 0 \leq a \leq d\}$$

when $n = 2d$, and in e^x times this span when $n = 2d + 1$ [1]. The identities

$$J'_0 = 2J_1, \quad J'_1 = 2J_0 - x^{-1}J_1$$

give a basis vector $\mathbf{F}$ satisfying

$$\mathbf{F}'(x) = (M_0 + x^{-1}M_1)\mathbf{F}(x), \quad (10)$$

where the eigenvalues of M_0 are

$$n, n-4, n-8, \dots, n-4d. \quad (11)$$

If $\mathbf{F} = \sum_{k \geq 0} \mathbf{f}_k x^k / k!$ and $\widehat{\mathbf{F}}(z) = \sum_{k \geq 0} \mathbf{f}_k z^k$, coefficient comparison gives

$$z(I - zM_0)\widehat{\mathbf{F}}'(z) = (M_1 + zM_0)\widehat{\mathbf{F}}(z) - M_1\mathbf{f}_0. \quad (12)$$

This system produces the specialized recurrences below. The asymptotic argument will instead use the exact Weyl integral carrier, so that its proof follows the same route as the accompanying formalization.

Put

$$\beta_n = \frac{n(n-1)}{4}, \quad \alpha_n = \frac{n + \sqrt{n^2 - 4}}{2},$$

and

$$C_n = \frac{n^{\beta_n}}{n!} \prod_{j=1}^n \frac{\Gamma(1+j/2)}{\Gamma(3/2)}. \quad (13)$$

Theorem 5.2 (Fixed-rank asymptotic). *For every fixed $n \geq 3$,*

$$b_{n,k} \sim K_n \alpha_n^k k^{-\beta_n}, \quad K_n = C_n \frac{\alpha_n}{n} \left(\frac{\sqrt{n^2 - 4}}{n} \right)^{\beta_n - 1}. \quad (14)$$

For $n = 2$, one has $b_{2,k} = 1$ exactly.

Proof. Regev's strip theorem, with the Mehta integral evaluated explicitly, gives

$$u_{n,k} \sim C_n n^k k^{-\beta_n} \quad (15)$$

[8, 5].

We now compare two exact Weyl moments. Put $d = \lfloor n/2 \rfloor$ and, for $\theta = (\theta_1, \dots, \theta_d) \in [0, \pi]^d$, write

$$\Delta_c(\theta) = \prod_{1 \leq i < j \leq d} (\cos \theta_j - \cos \theta_i)^2.$$

For $n = 2d$, set

$$S_n(\theta) = 2 \sum_{i=1}^d \cos \theta_i, \quad W_n(\theta) = \Delta_c(\theta) \prod_{i=1}^d (1 + \cos \theta_i), \quad \kappa_n = \frac{2^{d^2-d}}{d! \pi^d};$$

for $n = 2d + 1$, set

$$S_n(\theta) = 1 + 2 \sum_{i=1}^d \cos \theta_i, \quad W_n(\theta) = \Delta_c(\theta) \prod_{i=1}^d (1 - \cos \theta_i)(1 + \cos \theta_i), \quad \kappa_n = \frac{2^{d^2}}{d! \pi^d}.$$

Let $P_0(s) = 1$, $P_1(s) = s$, and $P_{k+2}(s) = sP_{k+1}(s) - P_k(s)$. The Weyl integral form of Gessel's identity, together with the finite transform in Equation (8), gives the exact identities

$$u_{n,k} = \kappa_n \int_{[0,\pi]^d} S_n(\theta)^k W_n(\theta) d\theta, \quad b_{n,k} = \kappa_n \int_{[0,\pi]^d} P_k(S_n(\theta)) W_n(\theta) d\theta.$$

The second identity is coefficientwise: the recurrence for P_k is exactly the finite Fibonacci transform induced by the rational substitution.

The positive maximum of S_n is n , attained at $\theta = 0$. With $\theta = x/\sqrt{k}$, uniformly for x in compact subsets of the positive orthant,

$$\left(\frac{S_n(x/\sqrt{k})}{n} \right)^k \rightarrow \exp\left(-\frac{\|x\|^2}{n} \right).$$

For $s > 2$, put $a(s) = (s + \sqrt{s^2 - 4})/2$. The recurrence gives

$$P_k(s) = \frac{a(s)^{k+1} - a(s)^{-k-1}}{\sqrt{s^2 - 4}}, \quad \frac{d}{ds} \log a(s) = \frac{1}{\sqrt{s^2 - 4}}.$$

Hence

$$\frac{P_k(S_n(x/\sqrt{k}))}{P_k(n)} \rightarrow \exp\left(-\frac{\|x\|^2}{\sqrt{n^2 - 4}} \right).$$

Under the same scaling, the Weyl weight and the Jacobian contribute $k^{-\beta_n} Q_n(x) dx$, where Q_n is a nonnegative homogeneous polynomial of degree $2\beta_n - d$. The elementary quadratic bounds for $1 - \cos t$ and the concavity of $\log a(S_n(\theta))$ near zero give an integrable Gaussian envelope, so dominated convergence applies. Outside a fixed neighborhood of zero and, in even rank, the all-minus point, $|S_n|$ is bounded away from n and the contribution is exponentially smaller. At the even-rank all-minus point, the factor $\prod_i (1 + \cos \theta_i)$ vanishes and makes that same-scale contribution lower order. This also covers the one-dimensional $n = 3$ case, where the negative endpoint has absolute scale $1 < 3$.

Define

$$I_n(c) = \int_{\mathbb{R}_+^d} e^{-c\|x\|^2} Q_n(x) dx.$$

The preceding limits and Equation (15) give

$$\kappa_n I_n(1/n) = C_n, \quad \frac{b_{n,k}}{P_k(n) k^{-\beta_n}} \rightarrow \kappa_n I_n(1/\sqrt{n^2 - 4}).$$

Since Q_n has degree $2\beta_n - d$, scalar change of variables yields

$$\frac{I_n(1/\sqrt{n^2 - 4})}{I_n(1/n)} = \left(\frac{\sqrt{n^2 - 4}}{n} \right)^{\beta_n}.$$

Finally,

$$P_k(n) \sim \frac{\alpha_n^{k+1}}{\sqrt{n^2 - 4}}.$$

Combining the last three displays gives

$$K_n = C_n \frac{\alpha_n}{n} \left(\frac{\sqrt{n^2 - 4}}{n} \right)^{\beta_n - 1},$$

which is Equation (14). □

The first dimension left open by Tenn admits a particularly explicit form. Let $C_r = \frac{1}{r+1} \binom{2r}{r}$.

Corollary 5.3 (The four-letter case). *Define*

$$c_{2r} = C_r C_{r+1}, \quad c_{2r+1} = C_{r+1}^2.$$

Then

$$b_{4,k} = \sum_{j=0}^{\lfloor k/2 \rfloor} (-1)^j \binom{k-j}{j} c_{k-2j}. \quad (16)$$

Moreover,

$$b_{4,k} \sim \frac{6(2+\sqrt{3})}{\pi} (2+\sqrt{3})^k k^{-3}. \quad (17)$$

Proof. The hook-length sum gives the bounded-height tableau formula

$$u_{4,2r} = C_r C_{r+1}, \quad u_{4,2r+1} = C_{r+1}^2.$$

Substitution in Theorem 4.2 gives Equation (16). Equation (13) gives $C_4 = 32/\pi$. Applying Theorem 5.2, with $\beta_4 = 3$, $\alpha_4 = 2 + \sqrt{3}$, and $\sqrt{4^2 - 4}/4 = \sqrt{3}/2$, gives Equation (17). $\square$

The next rank also admits a closed recurrence and a complete transferred asymptotic. Put

$$\alpha_5 = \frac{5 + \sqrt{21}}{2}.$$

Corollary 5.4 (The five-letter case). *Let $c_m = u_{5,m}$, with $c_0 = c_1 = 1$, $c_2 = 2$. For $m \geq 3$ these numbers satisfy*

$$\begin{aligned} (m+4)(m+6)c_m - (3m^2 + 17m + 15)c_{m-1} \\ - (m-1)(13m+9)c_{m-2} + 15(m-1)(m-2)c_{m-3} = 0. \end{aligned} \quad (18)$$

The ribbon number is

$$b_{5,k} = \sum_{j=0}^{\lfloor k/2 \rfloor} (-1)^j \binom{k-j}{j} c_{k-2j}, \quad (19)$$

and

$$b_{5,k} \sim \frac{1323\alpha_5}{8\pi} \alpha_5^k k^{-5}. \quad (20)$$

Proof. The height-five recurrence is the bounded-tableau recurrence of Bergeron and Gascon [1]; Equation (19) is Theorem 4.2. Equation (13) gives $C_5 = 9375/(8\pi)$. Now Theorem 5.2, with $\beta_5 = 5$ and local scale $\sqrt{21}/5$, gives

$$\frac{9375}{8\pi} \frac{\alpha_5}{5} \left(\frac{\sqrt{21}}{5} \right)^4 = \frac{1323\alpha_5}{8\pi},$$

which is Equation (20). $\square$

The next rank provides another explicit recurrence. Put $\alpha_6 = 3 + 2\sqrt{2}$.

Corollary 5.5 (The six-letter case). *Let $c_m = u_{6,m}$, with $c_0 = c_1 = 1$, $c_2 = 2$, and $c_3 = 4$. For $m \geq 4$,*

$$\begin{aligned} & (m+5)(m+8)(m+9)c_m - 4(5m^2 + 46m + 84)c_{m-1} \\ & - 4(m-1)(10m^2 + 58m + 33)c_{m-2} + 144(m-1)(m-2)c_{m-3} \\ & + 144(m-1)(m-2)(m-3)c_{m-4} = 0. \end{aligned} \quad (21)$$

Moreover,

$$b_{6,k} = \sum_{j=0}^{\lfloor k/2 \rfloor} (-1)^j \binom{k-j}{j} c_{k-2j}, \quad (22)$$

and

$$b_{6,k} \sim \frac{3 \cdot 2^{57/4} \alpha_6}{\pi^{3/2}} \alpha_6^k k^{-15/2}. \quad (23)$$

Proof. The differential equation for Y_6 in Bergeron and Gascon [1], after coefficient extraction, is precisely Equation (21). Equation (22) is Theorem 4.2. Finally, Equation (13) gives

$$C_6 = \frac{3}{4} \frac{6^{15/2}}{\pi^{3/2}}.$$

Theorem 5.2 has $\beta_6 = 15/2$ and local scale $\sqrt{32}/6 = 2\sqrt{2}/3$; simplifying

$$C_6 \frac{\alpha_6}{6} \left(\frac{2\sqrt{2}}{3} \right)^{13/2}$$

gives the constant in Equation (23). □

6 Stable involutions

Let I_k be the number of involutions of $[k]$. Robinson–Schensted gives a bijection between involutions and standard Young tableaux, because an involution has equal insertion and recording tableaux. Thus

$$I_k = \sum_{\lambda \vdash k} f^\lambda, \quad I_k = I_{k-1} + (k-1)I_{k-2}. \quad (24)$$

Corollary 6.1 (Stable range). *If $n \geq k$, then $b_{n,k}$ is the number a_k of involutions of $[k]$ having no cycle $(i, i+1)$. Equivalently, it is the number of matchings in the complement of the path P_k . In particular,*

$$a_k = \sum_{j=0}^{\lfloor k/2 \rfloor} (-1)^j \binom{k-j}{j} I_{k-2j}. \quad (25)$$

The initial values are

$$1, 1, 1, 2, 5, 13, 37, 112, 363, 1235, 4427, 16526, \dots$$

Proof. When $n \geq k$, every partition of $k-2j$ has at most n rows, so Theorem 4.2 and Equation (24) give Equation (25). To exclude adjacent transpositions directly, apply inclusion–exclusion to the path edges $(i, i+1)$. A nonempty intersection requires a matching of j path edges. Once those j transpositions are fixed, the remaining $k-2j$ labels support an arbitrary involution. There are $\binom{k-j}{j}$ such matchings, giving the same sum. □

The first finite-rank corrections below stability are also explicit. For $s, m \geq 0$, put

$$E_s(m) = \sum_{\substack{\lambda \vdash m \\ m - \lambda_1 \leq s}} f^\lambda. \quad (26)$$

Corollary 6.2 (Near-stable defect). *Fix $r \geq 1$ and let $k \geq r + 2$. Then*

$$a_k - b_{k-r,k} = \sum_{0 \leq j < r/2} (-1)^j \binom{k-j}{j} E_{r-2j-1}(k-2j). \quad (27)$$

For each fixed r , this defect is eventually a polynomial in k of degree $r - 1$ with leading coefficient $I_{r-1}/(r - 1)!$. The first four strips are

$$\begin{aligned} b_{k-1,k} &= a_k - 1, \\ b_{k-2,k} &= a_k - k, \\ b_{k-3,k} &= a_k - (k-1)(k-2), \\ b_{k-4,k} &= a_k - \frac{(k-2)(2k^2 - 8k + 3)}{3}. \end{aligned} \quad (28)$$

Proof. Subtract Theorem 4.2, with $n = k - r$, from the stable formula Equation (25). At the term $m = k - 2j$, the missing tableau shapes have more than $k - r$ rows. Conjugating their diagrams preserves f^λ and turns this condition into

$$\lambda_1 > k - r = m - (r - 2j),$$

so their total is exactly $E_{r-2j-1}(m)$. There is no defect once $2j \geq r$, proving Equation (27).

For fixed s , write every shape in Equation (26) as $(m - t, \nu)$ with $0 \leq t \leq s$ and $\nu \vdash t$. The hook-length formula shows that $f^{(m-t, \nu)}$ is eventually a polynomial in m of degree t with leading coefficient $f^\nu/t!$. Consequently $E_s(m)$ is eventually a polynomial of degree s with leading coefficient $\sum_{\nu \vdash s} f^\nu/s! = I_s/s!$. Thus the $j = 0$ term in Equation (27) supplies degree $r - 1$ with the stated leading coefficient, while every later term has smaller degree. Finally, direct hook-length evaluation gives

$$E_0(m) = 1, \quad E_1(m) = m, \quad E_2(m) = (m-1)^2, \quad E_3(m) = \frac{m(m-2)(2m-5)}{3},$$

in the ranges used above. Substitution proves Equation (28). $\square$

In particular, the stability threshold $n \geq k$ is sharp for every $k \geq 3$: the immediately preceding rank has exactly one fewer object.

The stable sequence is OEIS A170941 [7]. Substitution of the involution recurrence into Equation (25), followed by Pascal's identity, gives the useful recurrence

$$a_{k+1} = a_k + ka_{k-1} - a_{k-2} + a_{k-3} \quad (k \geq 3), \quad (29)$$

with $a_0 = a_1 = a_2 = 1$ and $a_3 = 2$.

For $n = 3$, the standard arc representation identifies the peakless Motzkin specialization with noncrossing involutions having no adjacent transposition.

7 The limiting proportion

Let Π_k be a uniformly random involution on $[k]$, and let

$$X_k = \#\{i : (i, i+1) \text{ is a cycle of } \Pi_k\}.$$

Theorem 7.1 (Poisson limit). *As $k \rightarrow \infty$,*

$$X_k \xrightarrow{d} \text{Poisson}(1).$$

Consequently

$$\frac{a_k}{I_k} \longrightarrow e^{-1}. \quad (30)$$

Proof. For fixed r , the falling factorial $(X_k)_r$ orders r distinct adjacent transpositions. They can occur simultaneously exactly when their path edges form a matching. Fixing them leaves an arbitrary involution on the remaining labels, so

$$\mathbb{E}[(X_k)_r] = r! \binom{k-r}{r} \frac{I_{k-2r}}{I_k}. \quad (31)$$

The classical involution asymptotic of Moser and Wyman [6] implies, for fixed r ,

$$\frac{I_{k-2r}}{I_k} \sim k^{-r}.$$

Since $r! \binom{k-r}{r} \sim k^r$, every fixed factorial moment in Equation (31) tends to 1. These are the factorial moments of $\text{Poisson}(1)$. The first moment makes the family tight. Moreover, for $M > r$,

$$\mathbb{E}[(X_k)_r \mathbf{1}_{\{X_k > M\}}] \leq \frac{\mathbb{E}[(X_k)_{r+1}]}{M - r},$$

so the bounded $(r+1)$ st factorial moments make $(X_k)_r$ uniformly integrable. Every subsequential limit therefore has all factorial moments equal to one and hence is $\text{Poisson}(1)$. This proves the distributional limit. Finally, evaluating the distribution functions at the continuity point $1/2$ gives $\mathbb{P}(X_k = 0) \rightarrow e^{-1}$. The event $X_k = 0$ is precisely the class counted by a_k , yielding Equation (30). $\square$

Equation (30) proves the limiting-proportion conjecture recorded in OEIS A170941. It also gives the stable ribbon asymptotic immediately from the Moser–Wyman formula.

Combining the fixed-rank and stable results distinguishes the two asymptotic regimes.

Corollary 7.2 (Finite-rank density and noncommuting limits). *For every fixed $n \geq 3$,*

$$\frac{b_{n,k}}{u_{n,k}} \sim \frac{\alpha_n}{n} \left(\frac{\sqrt{n^2 - 4}}{n} \right)^{\beta_n - 1} \left(\frac{\alpha_n}{n} \right)^k. \quad (32)$$

In particular,

$$\lim_{n \rightarrow \infty} \lim_{k \rightarrow \infty} \frac{b_{n,k}}{u_{n,k}} = 0, \quad \lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{b_{n,k}}{u_{n,k}} = e^{-1}. \quad (33)$$

Proof. Divide Equation (14) by Equation (15). Since $\alpha_n < n$ for every $n \geq 3$, the fixed-rank limit is zero. In the reverse order, fixed k enters the stable range once $n \geq k$, so the inner limit is a_k/I_k ; then apply Equation (30). $\square$

8 Further directions

The original all- n enumeration problem is resolved by Theorem 4.2; Theorems 4.3 and 5.2, together with Corollary 6.2, give the endpoint refinement, the complete fixed-rank leading asymptotic, and every fixed-distance correction below stability. Two natural directions remain. First, the stable involution model suggests that the finite-rank numbers should admit a direct positive arc-diagram interpretation with an n -dependent bounded statistic. The ordinary longest-decreasing-subsequence bound is not that statistic: the candidate count is 32 at $(n, k) = (4, 6)$, whereas Theorem 4.2 gives $b_{4,6} = 31$. The mixed defining/dual walk therefore records a different finite-rank quotient. Second, Equation (33) points to a uniform double-scaling problem between the fixed-rank and stable regimes. The proof here keeps n fixed before $k \rightarrow \infty$ and does not supply estimates uniform when both parameters grow.

Declaration of Generative AI and AI-assisted technologies in the writing process.

During the preparation of this work, the author used an author-developed local implementation of the Proof Engine framework for AI-assisted mathematical research. The implementation operates through replaceable AI-agent and harness components; the present work used OpenAI Codex within that implementation. The framework is described in *Proof Engine Infrastructure* [4]. The author reviewed and edited the resulting material and assumes full responsibility for the definitions, statements, proofs, citations, code, and exposition.

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